

Sugar Evidence: Read the Label

Opening

Learning objective

I can compare sugar values using the same label basis and order four supplied food labels from least to most sugar.

Success looks like

Find the sugar number, unit and label basis.

Compare values with the same unit and basis.

Order four labels and explain the limits of the evidence.

Arrival task

Arrival · 4 min across this stage

Standard

1. Read the arrow

Plant → caterpillar → bird. What does the arrow from plant to caterpillar show?

Help: Think about food and energy transfer.

2. Recall body jobs

Name two jobs that nutrients from food support.

Help: Think about energy, growth or repair.

3. Explain variety

Why does a balanced eating pattern include a range of foods?

Help: Different foods supply different nutrients.

4. Read all three

A label says 'of which sugars: 4 g per 100 g'. Give the number, unit and basis.

Help: Number = how much; unit = what measure; basis = in how much food.

Arrival answers

Arrival · 4 min across this stage

Standard

1. Read the arrow

The caterpillar gets food and energy by eating the plant. The arrow points in the direction of transfer.

2. Recall body jobs

Any two: supplying energy, growth, repairing tissues. Accept other correct nutrient functions.

3. Explain variety

Different foods provide different nutrients and amounts. A varied pattern helps meet different needs; no single food must do every job.

4. Read all three

Number: 4. Unit: g (grams). Basis: per 100 g of food.

Arrival task

Arrival · 4 min across this stage

Supported

1. Food chain

Plant → caterpillar → bird. Which is the plant that starts this chain?

Help: A producer makes its own food. Here, it is the plant.

2. Body jobs

Food helps us have energy and grow. Choose: energy only / growth only / both.

Help: There are two jobs in the question.

3. A varied pattern

A balanced eating pattern includes a range of foods. Choose: one food always / a range of foods.

Help: Balanced means varied over time.

4. Label clue

Teaching label: 'of which sugars: 4 g per 100 g'. Point to the sugar number.

Help: Find the number just before the first g.

Arrival answers

Arrival · 4 min across this stage

Supported

1. Food chain

The plant is the producer. The arrows show food and energy passing to the next living thing.

2. Body jobs

Both. Nutrients support energy needs, growth and repair.

3. A varied pattern

A range of foods. One food does not have to supply everything.

4. Label clue

4. The complete sugar value is 4 g per 100 g.

Arrival task

Arrival · 4 min across this stage

Stretch

1. Use food-chain evidence

Plant → caterpillar → bird. Explain why the arrows are not pointing towards what each animal hunts or eats.

Help: Follow the transfer of food and energy.

2. Separate the claims

'Food supplies energy' and 'This person eats badly'. Which is a statement about a body job?

Help: Keep the science about nutrients and evidence.

3. Limit a judgement

Can one food's sugar value describe a person's whole eating pattern? Explain.

Help: Consider the other foods and information missing.

4. Spot the mismatch

Can 8 g sugar per serving be compared directly with 8 g per 100 g? What is missing?

Help: A serving might not be 100 g.

Arrival answers

Arrival · 4 min across this stage

Stretch

1. Use food-chain evidence

The arrows show food and energy moving from the organism eaten to the consumer, not the direction the consumer travels to find food.

2. Separate the claims

'Food supplies energy' describes a body job. The second judges a person and is not supported by that nutrient fact.

3. Limit a judgement

No. One sugar value does not show all foods, nutrients, amounts or the pattern over time. Keep conclusions within the evidence.

4. Spot the mismatch

No, unless the serving is known to be 100 g. We need its mass and must put both values on the same basis.

Knowledge organiser: the whole label

Starter · 4 min across this stage

Reference

Label X / per 100 g	Amount
Carbohydrate	47 g
of which sugars	8 g

Number, unit and basis

8 = number: how much sugar

g = unit: grams measure mass

per 100 g = basis: food amount compared

Carbohydrate includes sugars. “Of which sugars” is the sugar part of that total. Evidence is information used to support a claim.

Knowledge organiser: equal amounts

Starter · 4 min across this stage

Reference

Label	Sugar / g per 100 g
P	6
Q	11
R	18
S	21

Worked example Z

2 g sugar per 20 g serving
 $5 \times 20 \text{ g} = 100 \text{ g}$ of food
 $5 \times 2 \text{ g} = 10 \text{ g}$ sugar
Z has 10 g sugar per 100 g.

With the basis matched: P, Z, Q, R, S.
“P has less sugar per 100 g than Q.”
Sugar alone cannot identify overall health.

Starter: which number matters?

Starter · 4 min across this stage

Nutrient / per 100 g	Label A	Label B
Carbohydrate	22 g	67 g
of which sugars	4 g	9 g

Point to a number you would use to compare sugar.
Why that number?

Fictional teaching labels. Nothing is tasted. Help: track the “of which sugars” row.

Number, unit and basis

I do 1 · 5 min across this stage

Label A / per 100 g	Amount
Carbohydrate	22 g
of which sugars	4 g
Protein	6 g
Salt	0.2 g

Sugars are already part of carbohydrate.
Do not add the two rows together.

Read the whole value

4
number

g
unit

per 100 g
basis

Reading and comparing labels

I do 1 · 5 min across this stage

Explanation 1 of 2

The sugar line

Find “of which sugars”. Read the number, unit and basis together: 4 g sugars per 100 g of food.

The same food amount

A has 4 g and B has 9 g sugars per 100 g. A has less sugar when equal food amounts are compared.

Reading and comparing labels

I do 1 · 5 min across this stage

Explanation 2 of 2

A different basis

E has 3 g sugars per 25 g serving. Four 25 g servings make 100 g. Four lots of 3 g give 12 g sugars per 100 g.

The limits of the numbers

These figures compare sugar only. They cannot identify the best whole food for everyone.

We do: order the labels

We do 1 · 5 min across this stage

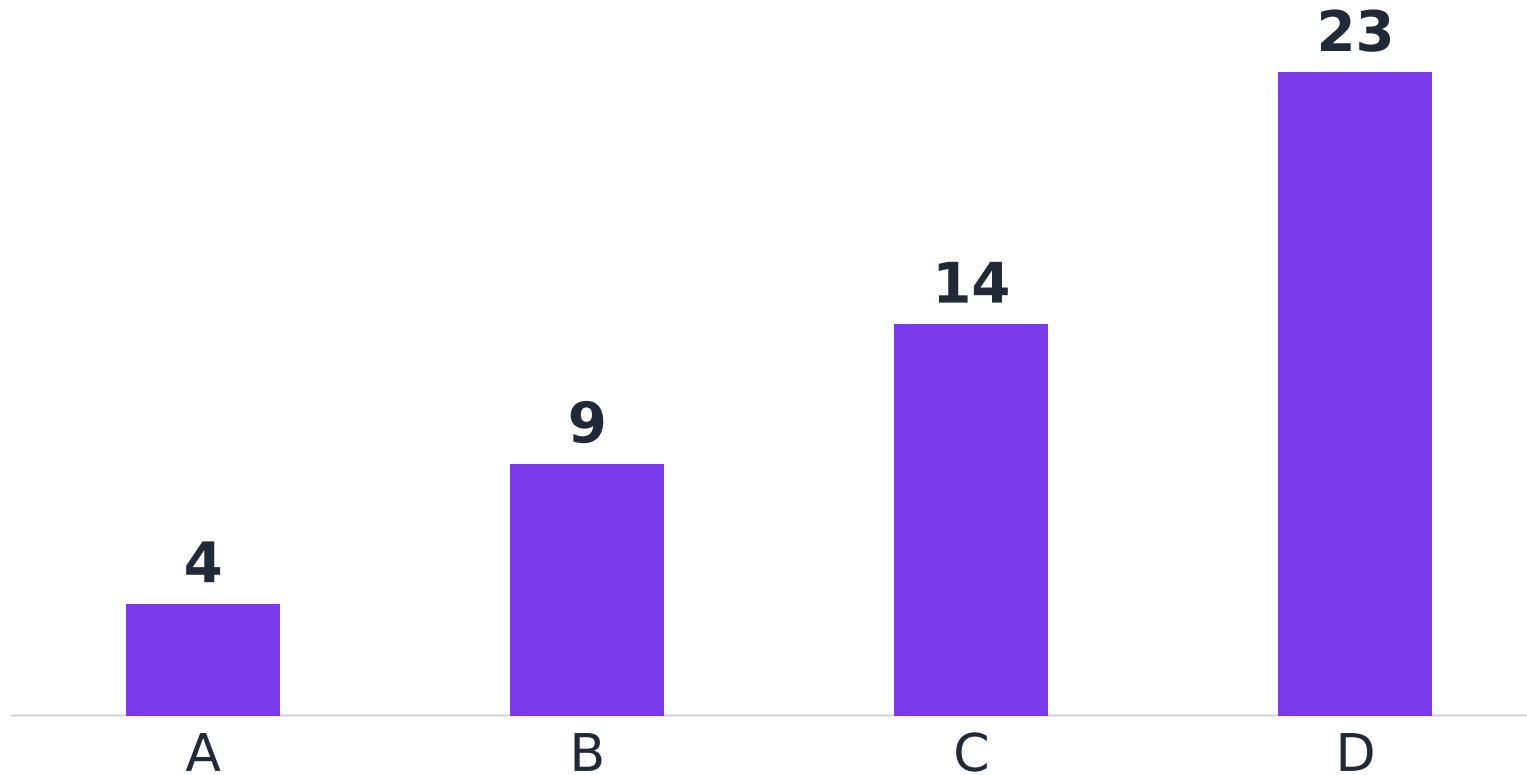
Label	Sugar / g per 100 g
C	14
A	4
D	23
B	9

Order from least to most sugar per 100 g.
Point, move the paper cards or direct an adult.

Our order: check the evidence

We do 1 · 5 min across this stage

Answer guidance



Sugar / g per 100 g

A, B, C, D

$4 < 9 < 14 < 23$

Check each
neighbouring pair.

A fair comparison

I do 2 · 4 min across this stage

Core check

Label	Food amount	Sugar
A	100 g	4 g
E	25 g serving	3 g

Are these the same food amounts?
Explain why comparing 4 and 3 directly is not fair.

Four equal servings

I do 2 · 4 min across this stage

Extension

Servings of E	Food mass	Sugar mass
1	25 g	3 g
2	50 g	6 g
3	75 g	9 g
4	100 g	12 g

$4 \times 25 \text{ g food} = 100 \text{ g food}$

$4 \times 3 \text{ g sugar} = 12 \text{ g sugar}$

We do: label detective

We do 2 · 4 min across this stage

of which sugars: 14 g per 100 g

Box the number. Circle the unit. Underline the basis.

Which sentence stays within the evidence?

A has less sugar per 100 g than D.

A must be the healthiest food for everyone.

Label detective: answer guidance

We do 2 · 4 min across this stage

Part	Meaning
14	Number: amount of sugar
g	Unit: grams
per 100 g	Basis: amount of food compared

“A has less sugar per 100 g than D” stays within the evidence.
The sugar values cannot establish the healthiest food for everyone.

You do: shared teaching labels

You do · 10 min across this stage

Evidence

Per 100 g	A	B	C	D
Carbohydrate / g	22	67	58	72
of which sugars / g	4	9	14	23

Challenge E: 3 g sugars per 25 g serving.
Four such servings weigh 100 g. All other information is unknown.

You do: use the evidence

You do · 10 min across this stage

Standard / 1 of 3

Question 1

Read these fictional teaching labels. Each is PER 100 g. A: carbohydrate 22 g; of which sugars 4 g. B: carbohydrate 67 g; of which sugars 9 g. C: carbohydrate 58 g; of which sugars 14 g. D: carbohydrate 72 g; of which sugars 23 g.

Help: Track the 'of which sugars' row. Keep the carbohydrate and sugar rows separate.

Question 2

Make a results row for each label. Copy the SUGAR value with its unit and basis; do not copy total carbohydrate.

Help: Use the headings Label | Sugar / g per 100 g. Check each copied value against its nutrient name.

You do: use the evidence

You do · 10 min across this stage

Standard / 2 of 3

Question 3

Order A–D from least to most sugar per 100 g.

Help: Check adjacent pairs: is each value no larger than the next? Recheck that all four use the same basis.

Question 4

Write or say one comparison: 'Label ____ has ____ g sugar per 100 g. This is less/more than ____.'

Help: Build the comparison from two complete readings. Include which label you are comparing it with.

You do: use the evidence

You do · 10 min across this stage

Standard / 3 of 3

Question 5

Explain one thing these sugar numbers cannot tell us about the whole food or eating pattern.

Help: Choose one missing piece of information. Explain why that missing information limits a wider judgement.

You may point, direct, draw, write or explain to an adult.

You do: answer guidance

You do · 10 min across this stage

Standard

The sugar values and order

A 4, B 9, C 14, D 23 g sugars per 100 g. The values 22, 67, 58 and 72 are total carbohydrate.

A comparison and a limit

B has 9 g sugar per 100 g, less than C at 14 g. Other nutrients, actual portions and the wider eating pattern are not all known.

You do: use the evidence

You do · 10 min across this stage

Supported / 1 of 3

Question 1

Use these fictional teaching cards. All give sugar in g per 100 g: A — 4 g; B — 9 g; C — 14 g; D — 23 g.

Help: Use a paper strip to track one card at a time. Say or point to the letter, the sugar number and 'g per 100 g' together.

Question 2

Put the four cards in order from LEAST sugar per 100 g to MOST. Point, move paper cards or direct an adult.

Help: Compare two numbers first. Place the smaller one towards LEAST, then find where each remaining card belongs.

You do: use the evidence

You do · 10 min across this stage

Supported / 2 of 3

Question 3

Complete: Label ____ has the least sugar per 100 g. Label ____ has the most.

Help: Look at the two ends of your own order. Use those card letters to fill the gaps.

Question 4

Choose the claim we can support: 'A has less sugar per 100 g than D' / 'A is the healthiest food for everyone'.

Help: Ask: does this sentence describe the sugar evidence, or claim to know everything about the food?

You do: use the evidence

You do · 10 min across this stage

Supported / 3 of 3

Question 5

Finish or point: These sugar numbers show ONE PART OF NUTRITION / THE WHOLE EATING PATTERN. Keep “g per 100 g” with each number.

Help: Look at what was measured. Think about whether these cards show every nutrient and everything a person eats.

You may point, direct, draw, write or explain to an adult.

You do: answer guidance

You do · 10 min across this stage

Supported

The order

A 4, B 9, C 14, D 23. All values are g sugars per 100 g. Least: A. Most: D.

The claim

A has less sugar per 100 g than D. These numbers show one part of nutrition, not the whole eating pattern.

You do: use the evidence

You do · 10 min across this stage

Stretch / 1 of 3

Question 1

Fictional labels, all PER 100 g: A — carbohydrate 22 g, sugars 4 g; B — carbohydrate 67 g, sugars 9 g; C — carbohydrate 58 g, sugars 14 g; D — carbohydrate 72 g, sugars 23 g.

Help: Identify the nutrient and reference amount before selecting numerical evidence.

Question 2

Extract the four sugar values and order them. Keep the unit and basis.

Help: Audit the completed order against the source labels; a correct number copied from the wrong row is still the wrong evidence.

You do: use the evidence

You do · 10 min across this stage

Stretch / 2 of 3

Question 3

Extra label E says 'sugars 3 g per 25 g serving'. Someone puts E before A because 3 is less than 4. Audit this decision.

Help: Compare the reference quantities before comparing the two bare numbers.

Question 4

Put E on a per-100 g basis before comparing. How many 25 g servings make 100 g? Then place E among A–D.

Help: Find the scale factor from 25 g of food to 100 g of food. Apply the same factor to the sugar amount.

You do: use the evidence

You do · 10 min across this stage

Stretch / 3 of 3

Question 5

Write one supported comparison from A–D and repair this claim: 'The lowest-sugar card proves the best food for everybody.'

Help: Separate a justified numerical comparison from a wider judgement. State exactly what the evidence establishes and what it leaves unknown.

You may point, direct, draw, write or explain to an adult.

You do: answer guidance

You do · 10 min across this stage

Stretch

E on the same basis

Four 25 g servings make 100 g. E has $3 \times 4 = 12$ g sugar per 100 g.
Order: A 4, B 9, E 12, C 14, D 23.

A comparison and a repair

A has 19 g less sugar per 100 g than D. The lowest sugar value does not prove the best food for everyone. More nutrition and context information is needed.

Exit ticket: two small checks

Exit · 4 min across this stage

Standard

Complete the comparison

Label B has ____ g sugars per 100 g.
That is less / more than C at 14 g per 100 g.

Finish the idea

These numbers compare sugar / every nutrient.
They do not tell us everything about the food.

Exit: answer guidance

Exit · 4 min across this stage

Standard

Check 1

9 g; less. The basis is per 100 g for both labels.

Check 2

Sugar. Accept an equivalent spoken completion.

Exit ticket: two small checks

Exit · 4 min across this stage

Supported

Which label has less sugar per 100 g?

B: 9 g per 100 g

C: 14 g per 100 g

What do these numbers tell us?

The amount of sugar

Everything about the food

Exit: answer guidance

Exit · 4 min across this stage

Supported

Check 1

B: 9 g sugars per 100 g, less than C at 14 g. Pointing to the complete B value is enough.

Check 2

The amount of sugar. These figures do not describe everything about the food.

Exit ticket: two small checks

Exit · 4 min across this stage

Stretch

Put E on the same basis

E has 3 g sugars per 25 g serving. Four servings make 100 g. E has 6 / 12 / 24 g sugars per 100 g. Compare with B at 9 g.

What else would you need to know?

Other nutrient amounts / The label font
Or say or write your own relevant example.

Exit: answer guidance

Exit · 4 min across this stage

Stretch

Check 1

12 g sugars per 100 g; more sugar than B at 9 g. Four 25 g servings make 100 g, so $3 \times 4 = 12$.

Check 2

Other nutrient amounts. Also accept a relevant own example, such as actual portion, other nutrients, individual needs or the wider eating pattern. The label font is not nutrition evidence.

Next lesson: Body Science Checkpoint

Exit · 4 min across this stage

What helped you read the whole sugar value today?

Number, unit and basis